

AI concepts

Q1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a technology that enables computers and machines to perform tasks that normally require human intelligence. These tasks include learning, problem-solving, understanding language, recognizing images, and making decisions. AI is used in virtual assistants, chatbots, recommendation systems, and self-driving vehicles.

Q2. Difference Between AI, Machine Learning, Generative AI, and Agentic AI Technology

Description

Artificial Intelligence (AI)

A broad field that enables machines to perform intelligent tasks.

Machine Learning (ML)

A subset of AI that allows computers to learn from data and improve without explicit programming.

Generative AI

AI that can create new content such as text, images, music, and videos.

Agentic AI

AI systems that can plan, make decisions, and perform tasks autonomously to achieve goals.

Example:

AI: Voice assistants.

ML: Spam email detection.

Generative AI: ChatGPT generating text.

Agentic AI: An AI assistant that can plan and complete multiple tasks automatically.

Q3. Explain Any Three Open-Source AI Models Available on Hugging Face

1. Llama 3

Llama 3 is a large language model developed by Meta. It can answer questions, generate text, summarize content, and assist with coding tasks.

2. Mistral 7B

Mistral 7B is a lightweight and efficient language model. It provides strong performance while requiring fewer computing resources.

3. FLAN-T5

FLAN-T5 is a language model developed by Google. It is trained to follow instructions and perform tasks such as translation, summarization, and question answering.

These models are freely available on Hugging Face and can be used for learning and development.

Q4. What is the Difference Between Traditional Programming and AI-Assisted Development?

Traditional Programming

AI-Assisted Development

Developers write all logic manually.

AI helps generate code and suggestions.

Rules are explicitly defined.

AI learns patterns from data.

Development can take more time.

Development can be faster.

Limited automation.

High level of automation and assistance.

Example: In traditional programming, a developer writes every step required for a calculator. In AI-assisted development, tools like GitHub Copilot can suggest code automatically, reducing development time.

Q5. Explain AI Hallucinations with One Example

AI hallucination occurs when an AI system generates information that sounds correct but is actually incorrect or fabricated. This happens because the model predicts likely answers rather than verifying facts.

Example: If an AI is asked about a book that does not exist, it may create a fake title, author, and summary instead of stating that the book is unavailable.

